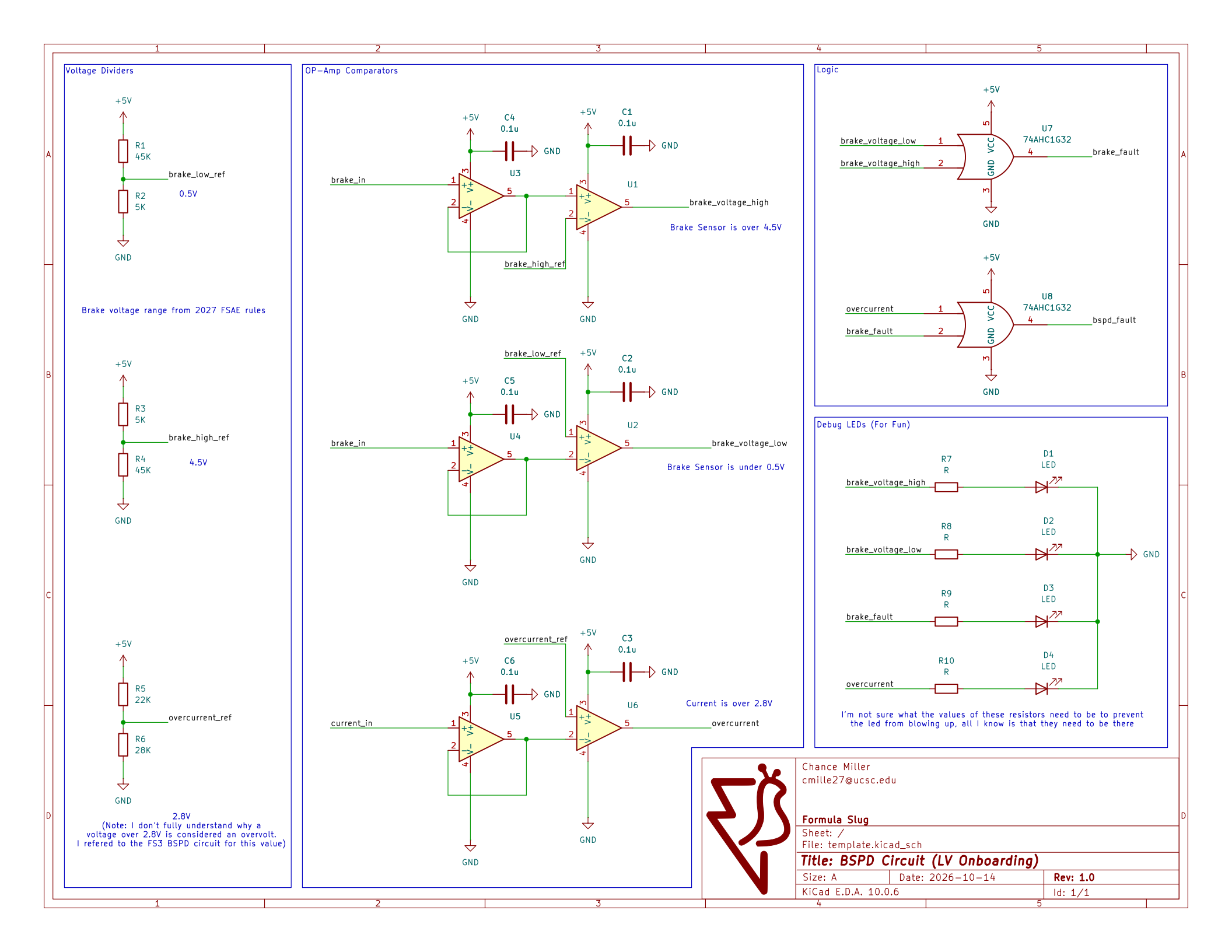

Voltage Dividers
+5V
R1
45K
brake_low_ref
0.5V
R2
5K
GND
Brake voltage range from 2027 FSAE rules
+5V
R3
5K
brake_high_ref
4.5V
R4
45K
GND
+5V
R5
22K
overcurrent_ref
R6
28K
GND
2.8V
(Note: I don't fully understand why a
voltage over 2.8V is considered an overvolt.
I refered to the FS3 BSPD circuit for this value)
OP-Amp Comparators
+5V
C4
0.1u
GND
U3
brake_in
brake_high_ref
+5V
C1
0.1u
GND
U1
brake_voltage_high
Brake Sensor is over 4.5V
GND
GND
brake_low_ref
+5V
C5
0.1u
GND
U4
brake_in
+5V
C2
0.1u
GND
U2
brake_voltage_low
Brake Sensor is under 0.5V
GND
GND
overcurrent_ref
+5V
C6
0.1u
GND
U5
current_in
+5V
C3
0.1u
GND
U6
Current is over 2.8V
overcurrent
GND
GND
Logic
+5V
U7
74AHC1G32
brake_voltage_low
brake_voltage_high
brake_fault
GND
GND
+5V
U8
74AHC1G32
overcurrent
brake_fault
bspd_fault
GND
GND
Debug LEDs (For Fun)
R7
R
D1
LED
brake_voltage_high
R8
R
D2
LED
brake_voltage_low
GND
R9
R
D3
LED
brake_fault
R10
R
D4
LED
overcurrent
I'm not sure what the values of these resistors need to be to prevent
the led from blowing up, all I know is that they need to be there
Chance Miller
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Formula Slug
Sheet: /
File: template.kicad_sch
Title: BSPD Circuit (LV Onboarding)
Size: A
Date: 2026-10-14
Rev: 1.0
KiCad E.D.A. 10.0.6
Id: 1/1